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6,80 6,54 6,90 7,00 6,53 6,93 6,79 6,53 6,52 6,82

6,80 6,91 6,90 7,00 6,76 6,93 6,79 6,53 6,52 6,81

6,71 6,90 6,90 7,00 6,53 6,93 6,79 6,53 6,72 6,92

6,80 6,54 6,90 6,93 6,93 6,93 6,79 6,53 6,52 6,54

$$n_1 = n_2 = 40$$

$$\bar{x}_1 = \frac{1}{40} \sum x_{1i} = \frac{2787}{40} \approx 59,68$$

$$s_1^2 = \frac{\sum x_i^2 - \frac{(\sum x_i)^2}{n}}{n-1} = \frac{720549 - \frac{2787^2}{40}}{39} \approx 29,99$$

$$s_1 = \sqrt{29,99} \approx 5,49$$

$$\bar{y} = \frac{270,85}{40} \approx 6,77$$

$$s_2^2 = \frac{7835,09 - \frac{220,85^2}{40}}{39} \approx 0,03$$

$$s_2 = \sqrt{0,03} \approx 0,17$$

$$H_0: \sigma_1^2 = \sigma_2^2$$

$$F_{\text{on}} = \frac{s_1^2}{s_2^2} = \frac{29,99}{0,03} = 833$$

$$k_1 = n_1 - 1 = 39$$

$$k_2 = n_2 - 1 = 39$$

$$\alpha = 0,05$$

$$\alpha/2 = 0,025 : F_{\text{кр}}(0,025; 39; 39) \approx 1,88$$

$$F_{\text{on}} > F_{\text{кр}} - H_0 \text{ не выполняется}$$

$$\sigma_0^2 = 0,1$$

$$H_0: \sigma_1^2 = \sigma_0^2 \approx 0,1$$

$$H_1: \sigma_1^2 > \sigma_0^2$$

$$\chi_{\text{on}}^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{39 \cdot 29,99}{0,1} \approx 9246,1$$

$$\chi_{\text{кр}}^2(\alpha; k) = \chi_{\text{кр}}^2(0,05; 39) \approx 57,52$$

$$\chi_{\text{on}}^2 > \chi_{\text{кр}}^2 - H_0 \text{ не выполняется. Преприятие } > 0,1$$

$$H_0: M(X_1) = M(X_2)$$

$$H_1: M(X_1) \neq M(X_2)$$

$$s_{\bar{x}-\bar{y}} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} = \sqrt{\frac{29,99}{40} + \frac{0,03}{40}} \approx \cancel{0,27}^{0,63}$$

$$t_{\text{un}} = \frac{\bar{x} - \bar{y}}{s_{\bar{x}-\bar{y}}} = \frac{54,68 - 6,77}{0,63} \approx 76,05$$

$$k = 39 + 39 = 78, \quad t_{\text{un}} \approx 7,99 \quad (\text{multi. comparison})$$

$t_{\text{un}} > t_{\text{crit}} - H_0$ is rejected

~~$$s^2 = \frac{\sum (x_i - \bar{x})^2 + \sum (y_i - \bar{y})^2}{n_1 + n_2 - 2} = \frac{29,99 + 0,03}{78} \approx 0,38$$~~

~~$$t_{\text{un}} = \frac{\bar{x} - \bar{y}}{s_{\bar{x}-\bar{y}}} = \frac{54,68 - 6,77}{\sqrt{0,38 \left(\frac{1}{40} + \frac{1}{40} \right)}} \approx 76,05$$~~

~~$$H_0: \mu = 0 \quad H_1: \mu \neq 0$$~~

~~$$\alpha = 0,05$$~~

~~$$k = 78, \quad t_{\text{crit}} \approx 2,02$$~~

~~$t_{\text{un}} > t_{\text{crit}} - H_0$ is rejected~~

$$H_0: \rho = 0 \quad H_1: \rho \neq 0$$

$$r = \frac{\overline{XY} - \overline{X} \cdot \overline{Y}}{s_1 \cdot s_2} = \frac{0,52}{4,99 \cdot 0,77} \approx 0,67$$

$$T_{cr} = \sqrt{\frac{r^2 (n-2)}{1-r^2}} \approx \sqrt{22,52} \approx 4,75$$

$$n = 2, \quad \alpha = 0,05 : t_{cr} = 2,02$$

$|T_{cr}| > t_{cr} : H_0$ ~~biyiklenmo~~

$$H_0: M(X_2) = 6,65 \quad H_1: M(X_2) \neq 6,65$$

$$s_{\bar{y}} = \frac{s_2}{\sqrt{n_2}} = \frac{0,77}{\sqrt{40}} \approx 0,123$$

$$t_{cr} = \frac{\bar{y} - a_0}{s_{\bar{y}}} = \frac{6,77 - 6,65}{0,123} \approx 1$$

$$n = 40 - 1 = 39 \quad \alpha = 0,05 \quad t_{cr} \approx 2,02$$

$|t_{cr}| > t_{cr} : H_0$ ~~biyiklenmo~~